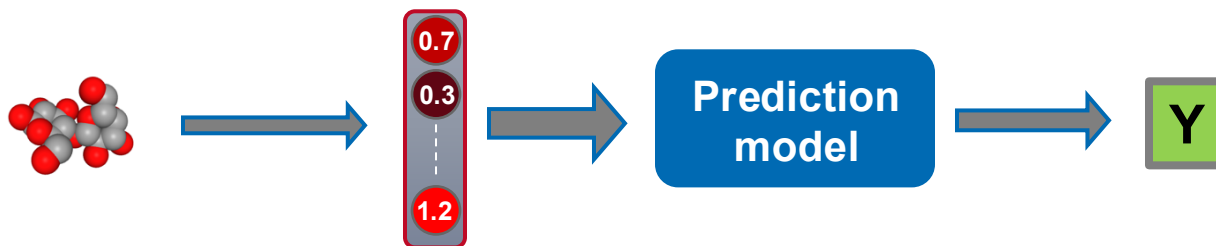
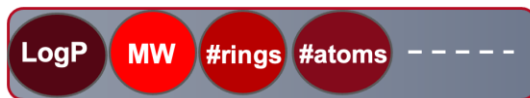


Traditional numerical representations for small molecules



■ Molecular Descriptors

- Examples for entries of molecular descriptors:
 - Number of atoms, number of bonds, molecular weight, number of rings, number of bonds, ...
 - Hydrophobicity (e.g. LogP), geometric descriptors (e.g. surface area)



- Many features can be calculated using the Python package RDKit
 - RDKit is an open-source cheminformatics software library
 - Has a *Descriptors* module with over 200 molecular descriptors

Molecular (expert-designed) Fingerprints

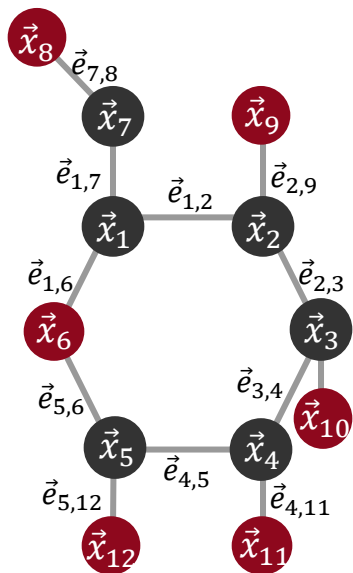
- Molecular (expert-designed) fingerprints:
 - Binary vectors representing the presence or absence of certain substructures in the molecule
 - Examples:
 - Graph-based:
 - Morgan Fingerprints (ECFPs)
 - RDKit Fingerprints
 - MACCS keys:
 - Each entry corresponds to a specific structural feature or characteristic of a molecule
 - Does the molecule contain a ring of size 4?
 - Are there fewer than 3 oxygen atoms present in the molecule?
 - 166-dimensional binary fingerprint
 - Can be calculated using the Python package RDKit



Morgan Fingerprints (ECFPs)

$$\vec{x}_i = \begin{pmatrix} \text{atom type} \\ \text{mass} \\ \text{valence} \\ \text{charge} \\ \dots \end{pmatrix} \quad \vec{e}_{i,j} = \begin{pmatrix} \text{bond type} \\ \text{part of ring} \\ \text{aromaticity} \\ \dots \\ \dots \end{pmatrix}$$

- Molecules are represented as graphs:
 - Atoms are interpreted as the nodes of a graph
 - Bonds are interpreted as the edges of a graph
- Calculating atom and bond feature vectors
- Updating each vector by using information about neighboring atoms and bonds
- Mapping all atom and bond representations from all iterations to a binary vector using hash functions

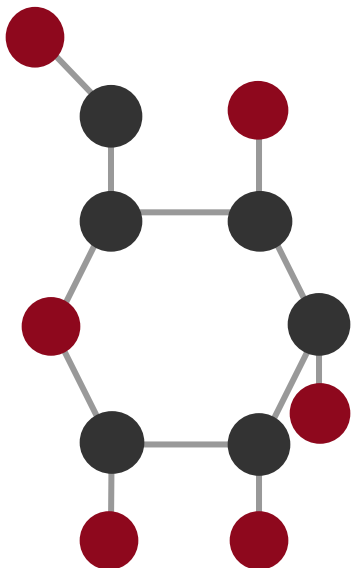


Morgan
Fingerprints

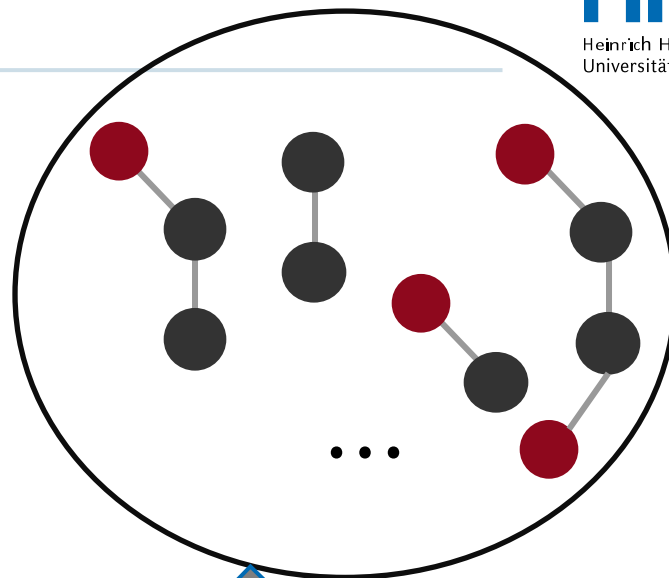


RDKit Fingerprints

**Interpret
molecule as graph**



**Identify all
subgraphs**



**RDKit
Fingerprint**

